

Beyond Raw Pace: Strategic Decision-Making in Formula 1

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- **Economic Problem**

- Formula 1 is a dynamic resource-allocation problem under uncertainty
- Teams trade off regulatory costs (engine/grid penalties) against strategic opportunities (Safety Car pit stops)

- **Strategic Gap**

- Car performance determines the baseline finishing order
- Strategy explains performance differences between teams of comparable pace ("strategic alpha")

- **Research Question**

- To what extent does race strategy create measureable value beyond raw vehicle performance?

Analytical Workflow

1 Data Integration

- Jolpica API, FastF1 and Open-Meteo
- Integrated dataset with 8,966 driver-race observations

2 Environment Modelling

- PCA + K-Means
- Three circuit archetypes used to control track heterogeneity

3 Causal Inference

- 2SLS: Safety-Car pit-stop effect
- Backdoor adjustment: Grid-penalty timing effect

4 Predictive Modelling

- Walk-forward XGBoost (14 folds)
- SHAP quantifies strategic predictive value

Results

Safety-Car Effect (2SLS)

- Estimate: **-10.1 pp**
- 95% CI: [-29.0, 8.8]
- No statistically significant causal effect

Penalty Timing (Backdoor)

- Estimate: **+12.8 pp**
- Robust across refutation tests
- Positive estimate ($p = 0.063$)

Predictive Performance

- Champion model: XGBoost (M4)
- ROC-AUC: **0.8656**
- 14-fold walk-forward validation

Strategic Alpha

- SHAP decomposition
- Strategic features explain
- **16.5%** of predictive variance

Main Finding

Strategy creates measurable value beyond raw vehicle performance. The strongest causal evidence is found for grid-penalty timing, while Safety-Car pitting shows no statistically significant causal effect.

Limitations

Data Coverage

- Strategy features available only from 2018 onwards
- Structural missingness in earlier seasons limits temporal coverage

Observed vs. Intended Strategy

- Grid penalties inferred from grid-position changes
- Team intent cannot be directly observed

Causal Identification

- Safety-Car IV estimated on only 72 races
- Wide confidence interval limits precision

Statistical Inference

- HC3 robust standard errors
- No race-level clustered inference

- **Main Finding**

- Car quality explains most race outcomes (61.7% of predictive variance)
- Strategy provides measurable value beyond raw vehicle performance

- **Economic Implication**

- Strategic decisions matter most where differences in car pace are small
- Midfield teams benefit more from strategic optimisation than front-runners

- **Future Research**

- Richer telemetry and team decision data would strengthen causal identification
- Larger Safety-Car samples would improve precision of the IV estimates

Backup: Model Comparison Table

Model	ROC-AUC	Strategic Lift
M1 (Baseline)	0.8485	-
M2 (Linear)	0.8504	+0.0019
M3 (Random Forest)	0.8619	+0.0134
M4 (XGBoost)	0.8656	+0.0171